Exprement 4: Relationships between Pixels: Neighbours, Connectivity and Distance Measures

1. Aim

To identify the 4-, diagonal and 8-neighbours of a pixel, study 4- and 8-connectivity, and calculate Euclidean, city-block and chessboard distances between two pixels using MATLAB.

2. Software Required

- Personal computer
- MATLAB R2020a or later
- Image Processing Toolbox

3. Program

Save the MATLAB program as exp04.m.

```
clc;

clear;

close all;

%% Part A: Neighbourhoods of a Pixel

A = magic(5);

disp('Matrix A =');

disp(A);

r = input('Enter row index (2 to 4): ');

c = input('Enter column index (2 to 4): ');

fprintf('Pixel value p(%d,%d) = %d\n', r, c, A(r,c));

N4 = [A(r-1,c), A(r+1,c), A(r,c-1), A(r,c+1)];

ND = [A(r-1,c-1), A(r-1,c+1), ...

    A(r+1,c-1), A(r+1,c+1)];

N8 = [N4, ND];

disp('4-neighbours N4 =');

disp(N4);

disp('Diagonal neighbours ND =');

disp(ND);
```

```
disp('8-neighbours N8 =');
disp(N8);
%% Part B: 4- and 8-Connectivity
BW = logical([1 1 0 0 0 0;
        1 1 0 0 1 0;
        0 0 1 0 1 0;
        0 0 0 1 0 0;
        0 1 0 0 0 1;
        0 1 0 0 0 1]);
CC4 = bwconncomp(BW, 4);
CC8 = bwconncomp(BW, 8);
fprintf('Components using 4-connectivity: %d\n', ...
   CC4.NumObjects);
fprintf('Components using 8-connectivity: %d\n', ...
   CC8.NumObjects);
figure('Name','Connectivity');
subplot(1,2,1);
imshow(BW);
title('Binary Image');
subplot(1,2,2);
imshow(label2rgb(bwlabel(BW,8)));
title('8-Connected Components');
%% Part C: Distance Between Two Pixels
I = imread('cameraman.tif');
figure('Name','Select Pixels');
imshow(I);
title('Click on Two Pixels');
[x,y] = ginput(2);
```

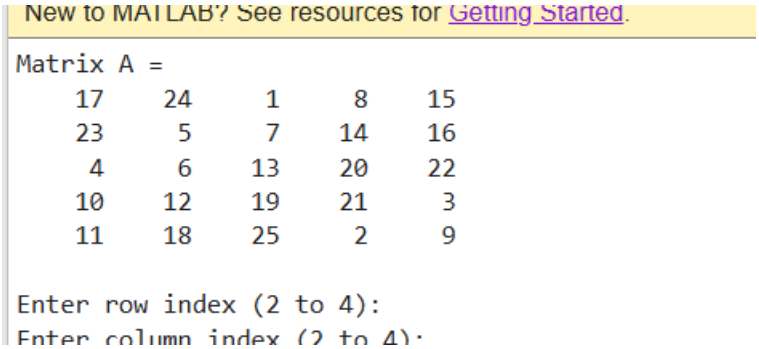
```
x = round(x);
y = round(y);
hold on;
plot(x,y,'r+-','MarkerSize',12,'LineWidth',1.5);
De = sqrt((x(1)-x(2))^2 + (y(1)-y(2))^2);
D4 = abs(x(1)-x(2)) + abs(y(1)-y(2));
D8 = max(abs(x(1)-x(2)), abs(y(1)-y(2)));
fprintf('p = (%d,%d), q = (%d,%d)\n', ...
   x(1),y(1),x(2),y(2));
fprintf('Euclidean distance De = %.4f\n', De);
fprintf('City-block distance D4 = %d\n', D4);
fprintf('Chessboard distance D8 = %d\n', D8);
%% Part D: Distance Maps
P = false(101);
P(51,51) = true;
figure('Name','Distance Maps');
subplot(1,3,1);
imshow(bwdist(P,'euclidean'),[]);
title('Euclidean Distance');
subplot(1,3,2);
imshow(bwdist(P,'cityblock'),[]);
title('City-block Distance');
subplot(1,3,3);
imshow(bwdist(P,'chessboard'),[]);
title('Chessboard Distance');
```

5. Procedure

1. Open MATLAB and set the Current Folder to the working directory.
2. Create a new script and save it as exp04.m.

3. Enter the program and run it by pressing F5.
4. Enter row and column indices between 2 and 4 to identify a pixel and its neighbours.
5. Observe the number of connected components for 4-connectivity and 8-connectivity.
6. Click two pixels in the displayed image to calculate their distances.
7. Observe the Euclidean, city-block and chessboard distance maps.
8. Record the calculated values in the observation table.

5. Output



8. Result

The 4-neighbours, diagonal neighbours and 8-neighbours of a pixel were identified. The 4- and 8-connectivity properties were studied, and Euclidean, city-block and chessboard distances between two pixels were calculated and visualized using MATLAB.